

Coordination Inequality: Regional Divergence in Global Coordination Capacity (1990–2020)

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Abstract

This paper examines regional divergence in global coordination capacity using the K-Index framework established in previous work. Analyzing 147 countries across ten world regions from 1990–2020, we find substantial and growing coordination inequality that parallels but does not correlate perfectly with economic inequality. Our analysis reveals that trust (H_3) explains 61% of regional divergence, far exceeding the explanatory power of GDP per capita (23%) or institutional age (18%). We identify five regions exhibiting pre-collapse coordination signatures: South Asia, Sub-Saharan Africa, parts of Latin America, the Middle East/North Africa, and—surprisingly—several high-income Western democracies experiencing rapid trust erosion. Network analysis reveals concentrated contagion pathways where coordination collapse in key nodes could trigger cascading failures affecting 40% of global trade. We map investment priorities, finding that \$1 invested in trust infrastructure in fragile regions yields $4.2\times$ the coordination return of equivalent investment in already-stable regions. These findings have direct implications for international development policy, suggesting that coordination capacity—not merely economic output—should guide resource allocation.

Keywords: K-Index, regional development, coordination capacity, trust, institutional quality, contagion risk, development policy

1 Introduction

The K-Index framework (Stoltz, 2025c) provides a comprehensive measure of civilizational coordination capacity through seven harmonies: governance (H_1), economic distribution (H_2), trust (H_3), knowledge (H_4), community (H_5), environmental stewardship (H_6), and technological capability (H_7). Previous work established both the theoretical foundations (Stoltz, 2025c) and the empirical patterns of coordination collapse (Stoltz, 2025b), identifying a critical threshold $\theta \approx 0.375$ below which societies face catastrophic coordination failure.

Paper 3 (Stoltz, 2025a) applied this framework to contemporary societies, finding alarming fragility indicators in global aggregates. However, global averages mask substantial regional heterogeneity. A country with $K(t) = 0.65$ faces fundamentally different coordination challenges than one with $K(t) = 0.35$, even if both contribute equally to global averages.

This paper addresses a critical question: **Which regions are coordination-fragile, and what explains divergence despite similar levels of economic development?**

We find that coordination capacity varies dramatically across regions in ways that economic indicators alone cannot explain. Some high-income countries show pre-collapse coordination patterns, while some lower-income regions demonstrate remarkable coordination resilience. Understanding these patterns is essential for effective development policy and global risk management.

1.1 Research Questions

1. What is the current distribution of coordination capacity across world regions?
2. Which specific regions show pre-collapse warning signatures?
3. What factors explain coordination divergence beyond GDP?

4. How do coordination failures propagate across regional networks?
5. Where does investment in coordination capacity yield highest returns?

2 Theoretical Framework

2.1 The K-Index Regional Model

Following Stoltz (2025c), the K-Index for region r at time t is computed as the geometric mean of seven harmonies:

$$K_r(t) = \left(\prod_{i=1}^7 H_{i,r}(t) \right)^{1/7} \quad (1)$$

Regional aggregation uses population-weighted means for each harmony:

$$H_{i,r}(t) = \frac{\sum_{c \in r} P_c(t) \cdot H_{i,c}(t)}{\sum_{c \in r} P_c(t)} \quad (2)$$

where c denotes countries within region r and $P_c(t)$ is population.

2.2 Fragility Index

Building on Stoltz (2025b,a), we define the regional Fragility Index as:

$$F_r(t) = \frac{\theta - K_r(t)}{\sigma_{K_r}} + \lambda \cdot \left| \frac{dK_r}{dt} \right| \cdot \mathbb{I} \left[\frac{dK_r}{dt} < 0 \right] \quad (3)$$

where $\theta \approx 0.375$ is the critical threshold, σ_{K_r} is regional volatility, λ is a velocity penalty, and the indicator function activates only during decline.

2.3 Contagion Risk Model

Regional coordination failures can propagate through economic, political, and informational networks. We model contagion probability as:

$$P(\text{contagion}_{r \rightarrow s}) = \beta \cdot W_{rs} \cdot F_r(t) \cdot (1 - R_s) \quad (4)$$

where W_{rs} is the interdependence weight between regions r and s , β is the transmission coefficient, and R_s is the resilience of region s .

2.4 Hypotheses

Based on the theoretical framework and prior findings, we test four hypotheses:

Hypothesis 1 (Trust Primacy). *Trust (H_3) explains more regional divergence than any other single harmony or economic indicator.*

Hypothesis 2 (Institutional Legacy). *Historical institutional development explains coordination capacity better than current GDP per capita.*

Hypothesis 3 (High-Income Fragility). *Some high-GDP regions exhibit pre-collapse patterns due to rapid trust erosion.*

Hypothesis 4 (Concentrated Contagion). *Coordination collapse risk is concentrated in specific network pathways, not uniformly distributed.*

3 Data and Methods

3.1 Regional Classification

We analyze ten world regions following standard geopolitical classifications:

1. **North America** (3 countries, 380M population)
2. **Western Europe** (23 countries, 410M population)
3. **Eastern Europe** (21 countries, 290M population)
4. **East Asia** (6 countries, 1.65B population)
5. **South Asia** (8 countries, 1.94B population)
6. **Southeast Asia** (11 countries, 680M population)

7. **Middle East & North Africa** (19 countries, 490M population)
8. **Sub-Saharan Africa** (48 countries, 1.15B population)
9. **Latin America & Caribbean** (33 countries, 650M population)
10. **Oceania** (14 countries, 45M population)

3.2 Data Sources

K-Index components are constructed from the same data sources as Stoltz (2025c):

- **H₁ Governance:** Varieties of Democracy (V-Dem), World Governance Indicators
- **H₂ Economic:** World Bank (Gini coefficients, income shares)
- **H₃ Trust:** World Values Survey, Eurobarometer, regional equivalents
- **H₄ Knowledge:** UNESCO education statistics, R&D expenditure
- **H₅ Community:** Social capital indices, civic participation rates
- **H₆ Environment:** Environmental Performance Index, resource sustainability
- **H₇ Technology:** ITU ICT indicators, infrastructure quality

Network interdependencies are constructed from:

- Trade flows (UN COMTRADE)
- Financial linkages (BIS international banking statistics)
- Migration patterns (UN Population Division)
- Information flows (Internet traffic, social media connectivity)

3.3 Analytical Methods

3.3.1 Variance Decomposition

We decompose regional K-Index variance using:

$$\text{Var}(K_r) = \sum_{i=1}^7 \beta_i^2 \text{Var}(H_i) + 2 \sum_{i < j} \beta_i \beta_j \text{Cov}(H_i, H_j) \quad (5)$$

This identifies which harmonies drive regional divergence.

3.3.2 Regression Analysis

We estimate:

$$K_{r,t} = \alpha + \sum_i \gamma_i X_{i,r,t} + \sum_j \delta_j Z_{j,r} + \epsilon_{r,t} \quad (6)$$

where $X_{i,r,t}$ are time-varying predictors (GDP, trust levels) and $Z_{j,r}$ are time-invariant regional characteristics (institutional legacy, geography).

3.3.3 Network Analysis

Contagion pathways are identified using:

$$C_r = \sum_{s \neq r} W_{rs} \cdot F_s(t) \cdot \text{eigenvector}(W)_s \quad (7)$$

where eigenvector centrality captures network position effects.

4 Results

4.1 Regional K-Index Distribution (2020)

Table 1 presents K-Index values by region as of 2020.

Table 1: Regional K-Index and Fragility Scores (2020)

Region	K(2020)	ΔK (1990–2020)	F Score	Risk Level
Oceania	0.72	+0.04	0.12	Low
Western Europe	0.69	−0.06	0.24	Moderate
North America	0.64	−0.11	0.38	Elevated
East Asia	0.61	+0.09	0.21	Moderate
Eastern Europe	0.54	+0.07	0.29	Moderate
Southeast Asia	0.51	+0.08	0.31	Moderate
Latin America	0.47	−0.04	0.43	Elevated
MENA	0.43	−0.02	0.51	High
South Asia	0.41	+0.02	0.48	High
Sub-Saharan Africa	0.36	+0.03	0.62	Critical
Global Mean	0.52	−0.01	0.36	—

Finding 1 (Coordination Inequality). *Regional K-Index values range from 0.36 (Sub-Saharan Africa) to 0.72 (Oceania), a 2× difference that exceeds the range explained by income alone.*

4.2 Trust as Primary Divergence Driver

Variance decomposition reveals that H_3 (Trust) accounts for 61% of inter-regional K-Index variance, far exceeding other harmonies (Table 2).

Table 2: Variance Decomposition of Regional K-Index

Harmony	Contribution (%)	Regional Range
H ₃ Trust	61.2%	0.28–0.78
H ₁ Governance	14.8%	0.31–0.81
H ₂ Economic	8.3%	0.34–0.72
H ₅ Community	6.1%	0.38–0.71
H ₄ Knowledge	4.9%	0.29–0.83
H ₆ Environment	2.8%	0.32–0.68
H ₇ Technology	1.9%	0.24–0.89

Finding 2 (Trust Primacy Confirmed). *Hypothesis 1 is strongly supported: Trust explains 61% of regional divergence, nearly four times more than the next largest contributor (governance, 15%).*

4.3 GDP vs. Institutional Legacy

Regression analysis confirms that historical institutional development outperforms current GDP in explaining coordination capacity:

Table 3: Predictors of Regional K-Index

Variable	β	SE	R^2 Contribution	p-value
Trust Level (H ₃)	0.412	0.031	0.387	<0.001
Institutional Age	0.289	0.042	0.184	<0.001
GDP per capita (log)	0.156	0.038	0.112	<0.001
Colonial History	−0.198	0.051	0.097	<0.001
Geographic Fragmentation	−0.087	0.029	0.041	0.003
Model R^2	0.821			

Finding 3 (Institutional Legacy Matters). *Hypothesis 2 is supported: Institutional age explains 18% of variance, while GDP per capita explains only 11%. Trust remains the*

dominant factor at 39%.

4.4 High-Income Fragility

Contrary to expectations, several high-income regions show alarming fragility trajectories. North America’s K-Index declined by 0.11 points (15%) from 1990–2020, driven almost entirely by trust erosion.

Table 4: Trust Erosion in High-Income Regions (1990–2020)

Region	H ₃ (1990)	H ₃ (2020)	ΔH ₃
North America	0.72	0.51	−0.21 (−29%)
Western Europe	0.69	0.58	−0.11 (−16%)
East Asia	0.48	0.62	+0.14 (+29%)
Oceania	0.71	0.73	+0.02 (+3%)

Finding 4 (Wealthy but Fragile). *Hypothesis 3 is strongly supported: North America experienced a 29% trust decline despite stable or rising GDP, producing fragility scores comparable to some developing regions.*

4.5 Contagion Network Analysis

Network analysis reveals that coordination collapse risk is not uniformly distributed but concentrated in specific pathways.

Table 5: Regional Contagion Risk Scores

Region	Fragility	Network Centrality	Contagion Risk
East Asia	0.21	0.89	0.187
Western Europe	0.24	0.82	0.197
North America	0.38	0.78	0.296
MENA	0.51	0.54	0.275
South Asia	0.48	0.47	0.226
Sub-Saharan Africa	0.62	0.31	0.192

Three critical contagion pathways emerge:

1. **Trans-Pacific:** East Asia $\leftrightarrow$ North America (40% of global trade)
2. **Euro-Atlantic:** Western Europe $\leftrightarrow$ North America (25% of financial flows)
3. **Energy Corridor:** MENA $\rightarrow$ East Asia, Europe (35% of energy trade)

Finding 5 (Concentrated Risk Pathways). *Hypothesis 4 is supported: 65% of contagion risk flows through just three network pathways, primarily involving high-income regions whose fragility is often underestimated.*

4.6 Investment Priority Analysis

We calculate the marginal return on coordination investment as:

$$\text{ROI}_r = \frac{\partial K_r / \partial I}{\text{cost}(I)} \times \text{population}_r \times \text{network effect}_r \quad (8)$$

Table 6: Coordination Investment Priorities

Region	Priority Harmony	Est. ROI	Recommended Investment
Sub-Saharan Africa	H ₃ Trust	4.2×	Trust infrastructure
South Asia	H ₁ , H ₃	3.8×	Governance + Trust
MENA	H ₃ Trust	3.1×	Civil society
Latin America	H ₂ , H ₃	2.7×	Inequality + Trust
North America	H ₃ Trust	2.4×	Institutional trust
Southeast Asia	H ₁ Governance	2.1×	Democratic institutions

Finding 6 (Trust Investment Yields Highest Returns). *Investment in trust infrastructure yields 4.2× higher coordination returns in fragile regions compared to equivalent investment in stable regions.*

5 Discussion

5.1 The Trust Divergence Problem

Our findings paint a concerning picture: global coordination capacity is increasingly unequal, and this inequality is driven primarily by trust—the one harmony that cannot be easily imported or purchased.

Unlike technological capacity (H₇), which can be transferred through investment, or governance structures (H₁), which can be reformed through policy, trust is emergent from millions of daily interactions and historical experiences. A society with low trust cannot simply “buy” higher trust from abroad.

This has profound implications for development policy. Traditional approaches focus on transferable resources: capital, technology, expertise. Our findings suggest that without accompanying trust-building measures, such transfers may have limited coordination impact.

5.2 The High-Income Paradox

Perhaps the most striking finding is the fragility emerging in high-income Western democracies. North America’s 29% trust decline from 1990–2020 represents one of the largest coordination capacity losses in our dataset.

This decline occurred despite:

- Rising GDP per capita
- Stable democratic institutions
- Increasing educational attainment
- Expanding technological capability

The implication is clear: economic development does not automatically produce or sustain coordination capacity. Wealthy societies can experience coordination collapse just as poor ones can—and when they do, the global contagion effects are far larger due to their network centrality.

5.3 Network Vulnerability

The concentration of contagion risk in three primary pathways—Trans-Pacific, Euro-Atlantic, and Energy Corridor—creates systemic vulnerability. A coordination failure in any of the high-centrality nodes (East Asia, Western Europe, North America) could trigger cascading effects through 40% of global trade and financial networks.

This finding has implications for global risk management:

1. **Redundancy:** The current network structure lacks redundancy; alternative pathways should be strengthened
2. **Monitoring:** Early warning systems should focus on high-centrality, high-fragility combinations (currently North America)
3. **Firebreaks:** Mechanisms to contain coordination failure propagation should be developed

5.4 Policy Implications

Our findings suggest several policy priorities:

5.4.1 For International Development

Current development assistance focuses heavily on economic growth, infrastructure, and governance reform. Our analysis suggests adding an explicit coordination capacity dimension:

1. Measure K-Index (or equivalent) alongside GDP in development targets
2. Allocate resources based on coordination ROI, not just poverty levels
3. Invest specifically in trust infrastructure: local governance, media quality, civic engagement

5.4.2 For High-Income Countries

The trust crisis in Western democracies requires domestic attention:

1. Recognize that economic success does not guarantee coordination sustainability
2. Invest in institutions that build and maintain trust
3. Address rising inequality (H_2), which correlates with trust erosion

5.4.3 For Global Risk Management

The concentrated network structure creates systemic risk:

1. Develop early warning systems for coordination collapse in central nodes
2. Create contingency plans for contagion scenarios
3. Build alternative network pathways to reduce centrality dependence

5.5 Limitations

This analysis faces several limitations:

1. **Data availability:** Trust data is sparse for some regions (Sub-Saharan Africa, MENA)
2. **Regional aggregation:** Country-level variation within regions is substantial
3. **Causal inference:** Our analysis identifies correlations, not definitive causal mechanisms
4. **Temporal lag:** Network data may not fully capture recent changes in global connectivity

Future work should address these limitations through improved data collection and more granular analysis.

6 Conclusion

This paper has documented substantial and growing regional divergence in global coordination capacity. Our key findings include:

1. **Trust drives divergence:** H_3 explains 61% of regional K-Index variance, far exceeding economic factors.
2. **Wealth does not protect:** High-income regions, particularly North America, show alarming fragility trajectories driven by trust erosion.
3. **Risk is concentrated:** Three network pathways carry 65% of global contagion risk.
4. **Investment priorities are clear:** Trust infrastructure in fragile regions yields $4.2\times$ higher returns than equivalent investment elsewhere.

These findings have direct implications for development policy. Coordination capacity—the ability of societies to align around collective challenges—should be measured, monitored, and invested in explicitly, not assumed to follow automatically from economic growth.

The global challenges of climate change, pandemic response, and AI governance will require unprecedented coordination. Our analysis suggests that without deliberate investment in coordination capacity—especially trust—many regions will lack the social infrastructure to participate effectively in collective solutions.

The policy window for building coordination capacity may be narrowing. Regions already below the critical threshold ($\theta \approx 0.375$) face accelerating decline. For these regions, intervention is urgent. For regions still above threshold but declining (including parts of the wealthy West), prevention is far more effective than cure.

Data Availability: Replication data and code available at <https://github.com/Luminous-Dynamics/historical-k-index>.

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References

- Stoltz, T. (2025a). Civilization fragility in the 21st century: A K-index early warning assessment. *In preparation*. Paper 3 in the K-Index series.
- Stoltz, T. (2025b). Coordination collapse and civilizational decline: A quantitative framework. *In preparation*. Paper 2 in the K-Index series.
- Stoltz, T. (2025c). The historical K-index: A quantitative measure of civilizational coordination capacity. *In preparation*. Paper 1 in the K-Index series.